

STEAMROOT · DESIGN PRECIS

SteamRoot design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update; record Amish's 2026-09-25 decisions; replace first-order numbers with STR-CAL-001 results; water-seal pot, two hoods and alarms added; regulatory ruling marked as an open prerequisite
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

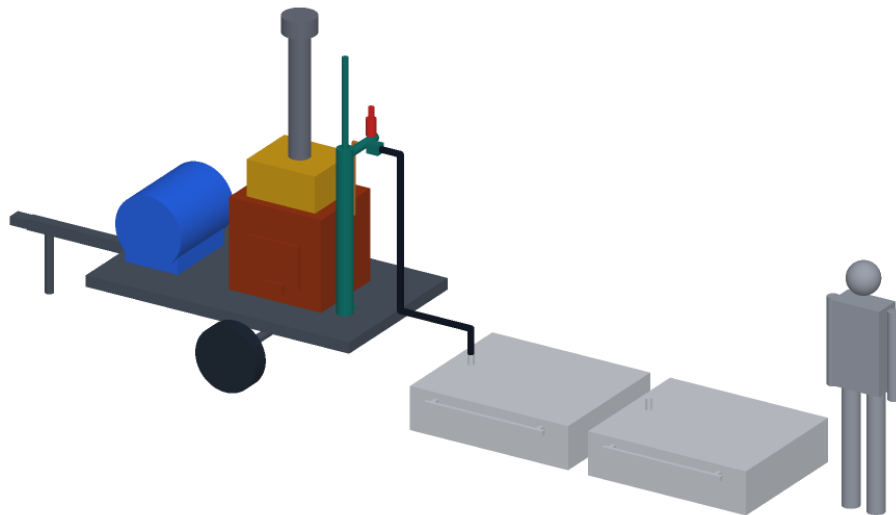
SteamRoot design precis

SteamRoot is a towable, wood-fired steam generator that never holds pressure. A pump pushes water once through a large-bore stainless coil wound around the fire, the steam flows at close to atmospheric pressure through a hose to one of two insulated hoods pressed onto the soil, and a flue-gas economizer preheats the feed water. The TRL 3 calculation (STR-CAL-001) confirms the core safety idea on paper: 30 kg/h of steam with under 7 L of water in the heated section and a header pressure of about 0.03 bar. It also shows that the concept misses several performance targets: about 54 % fuel-to-steam efficiency (target 65 %), about 1.9 m³/h at 15 cm (target 2 m³/h), 536 kg loaded (target 500 kg) and \$1,915 in parts (budget \$1,800).

Safety: SteamRoot is a concept for review, not a design to build. It combines open fire, carbon monoxide, and steam that causes severe burns in under a second. Local boiler and pressure vessel rules may apply even at low pressure. **A written ruling from the local boiler authority is an open prerequisite:** Amish approved getting it before TRL 3 detail work, and it has not been obtained. See the Safety section before any further work.

SteamRoot

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. TRL 3 model in the working position: trailer, generator and two hoods on a bed, next to a 1.75 m person.

■ How it works

1. **Feed.** A 12 V diaphragm pump draws water from a 125 L tank and meters 0.5 L/min (30 kg/h) into the economizer at a fixed rate. A float switch warns when the tank runs low.
2. **Preheat.** The economizer coil sits in the flue gas above the firebox and warms the feed water from about 15 °C to about 85 °C.
3. **Boil.** The warm water enters the bottom of a monotube coil wound around the fire bed inside the fiber-lined firebox. It boils as it rises through 11 turns and leaves the top of the coil as steam. The coil holds under 6 L of water. A thermocouple on the coil outlet sounds an alarm if the outlet overheats, which is the sign of a coil running dry.
4. **Vent and separate.** The coil discharges into a steam header 1.6 m above the ground. The header is always open to the atmosphere through a water-seal pot: a dip leg reaches 1 m below the water line, so steam can only build about 0.094 bar gauge before it bubbles through the seal and out of a vent pipe that ends 2.3 m above the ground, above head height. The seal cannot stick or be adjusted, and it is never valved off. A certified relief valve on the header is the second line of protection.
5. **Deliver.** Steam flows through a 6 m, 25 mm steam hose to a hood. A three-way diverter at the header sends steam to the hose or to the vent and has no closed position.
6. **Pasteurize.** Each hood is a 1.2 x 1.0 m insulated open-bottom pan with a 60 mm soil skirt. Steam from a perforated manifold condenses in the soil and drives a hot front downward. While one hood heats, the other holds its soil at temperature. To change over, the operator diverts steam to the vent, moves the hose coupling to the other hood and diverts steam back.

SteamRoot: energy flow at 30 kg/h steam, 15 cm depth (all values calculated estimates, STR-CAL-001)

CONCEPT, NOT FOR FABRICATION

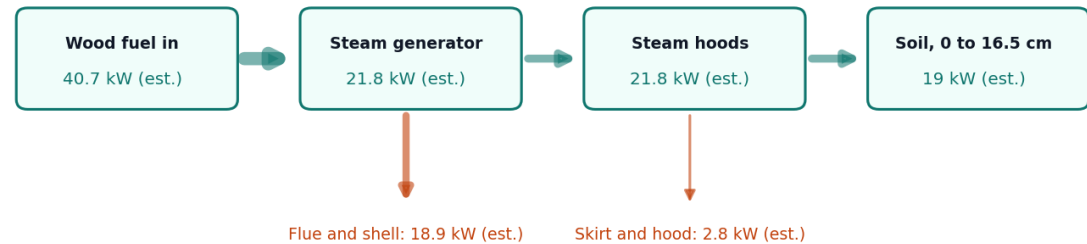


Figure 2. Energy flow at the design point of 30 kg/h of steam and 15 cm depth. All values are calculated estimates from STR-CAL-001.

■ Main components

Item numbers match the exploded view, the general arrangement drawing STR-DWG-002 and bom/bom.csv. Items 13 to 16 are not drawn.

Table 1. Main components.

#	Component	Choice	Notes
1	Trailer frame and drawbar	Used single-axle garden or utility trailer, deck 2.0 x 1.2 m	Rated 750 kg or more
2	Wheels	Trailer wheels, about 460 mm diameter	Included with the trailer
3	Feed water tank	125 L HDPE drum with strainer and sight tube	4.2 h of steaming
4	Feed pump	12 V diaphragm pump with needle valve and flow meter	Its 3 bar shut-off pressure also caps coil inlet pressure if the coil blocks
5	Firebox	700 x 600 x 650 mm steel box, 50 mm ceramic fiber lining, grate, latched door, air damper	About 41 kW firing. A castable lining would add about 200 kg
6	Monotube steam coil	316 stainless tube, 25.4 mm OD x 1.65 mm wall, 14.5 m in 11 turns on 420 mm	Decided by Amish, 2026-09-25
7	Flue-gas economizer	8 m of 12.7 mm stainless tube in a box above the firebox, with condensate drain	Flue condensate is acidic (dew point about 45 °C)
8	Chimney and spark arrestor	150 mm flue, outlet 2.4 m above ground, 6 mm mesh	Keeps flue gas above head height
9	Steam header and water-seal vent	DN50 header at 1.6 m, 114 mm seal pot with 1 m dip leg, vent to 2.3 m, three-way diverter, quick coupling	Primary pressure limit. Decided by Amish, 2026-09-25
10	Certified relief valve	ASME Section IV steam safety valve, 15 psi (1.03 bar) set, 3/4 in	Secondary protection only
11	Steam hose	EPDM saturated-steam hose, 25 mm bore, 6 m, ground-joint couplings, whip checks	Steam rated, not hot-water rated
12	Steam hoods (2)	1.2 x 1.0 x 0.25 m aluminum double-skin pans, 40 mm mineral wool, soil skirt, handles on standoffs	Two hoods used alternately. Decided by Amish, 2026-09-25
13	Battery	12 V 20 Ah sealed lead acid	Not drawn
14	Instruments	Steam gauge, four type K thermocouples and reader	Soil probes at 15 cm; not drawn
15	Safety kit	Personal CO alarm, fire extinguisher, gloves, face shield	Not drawn
16	Feed and coil alarms	Coil outlet temperature alarm, tank low-level switch	Decided by Amish, 2026-09-25; not drawn

SteamRoot: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Trailer frame and drawbar
- 2 Wheels
- 3 Feed water tank, 125 L
- 4 Feed pump
- 5 Firebox, fiber lined
- 6 Monotube steam coil
- 7 Flue-gas economizer
- 8 Chimney and spark arrestor
- 9 Steam header and water-seal vent
- 10 Certified relief valve
- 11 Steam hose
- 12 Steam hoods, 1.2 x 1.0 m (2)

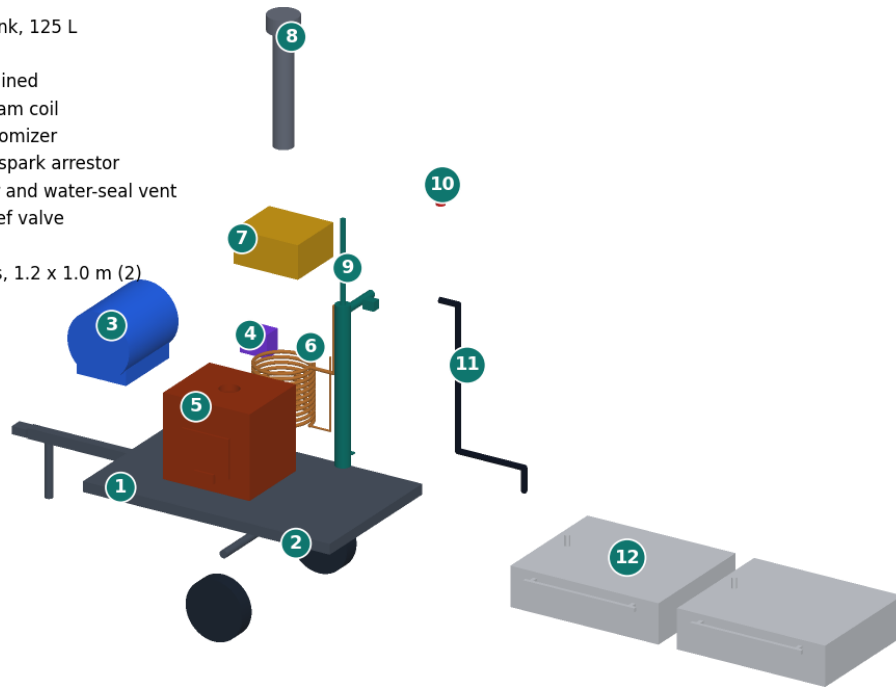


Figure 3. Exploded view with item numbers matching the BOM.

■ TRL 3 numbers

All values come from STR-CAL-001 (docs/04-calcs/01-sizing.md , script docs/04-calcs/sizing.py). They are paper estimates; none is tested.

Table 2. Key results.

Quantity	Value	Requirement
Steam output	30 kg/h, firing 40.7 kW, firebox gas 520 °C	R3 met
Heat into the water	21.78 kW (coil 19.34 kW, economizer 2.44 kW)	
Fuel to steam efficiency	53.5 % (range 44 % to 59 % at $\lambda = 2.0$ to 2.5 with $F = 0.40$ to 0.60; see STR-CAL-001, Table 5)	R5 not met
Stack temperature	448 °C (295 °C would be needed for 65 %)	
Wood use	10.1 kg/h at 14.5 MJ/kg	
Steam per m ² at 15 cm	15.6 kg/m ² (front to 16.5 cm)	
Treatment rate, two hoods	1.88 m ² /h at 15 cm; 3.41 m ² /h at 5 cm	R2 not met
Wood per m ² at 15 cm	5.4 kg/m ²	R4 not met
Header pressure, normal	0.033 bar gauge	
Coil inlet pressure, normal	0.110 bar gauge	R6 not met (10 % over)
Water seal limit	0.094 bar gauge	R6 open vent met by design
Water in heated section, flooded	6.98 L	R7 met
Steaming per fill	4.2 h	R8 met
Relief valve orifice needed	6.2 mm at 30 kg/h; 14.1 mm for a dry-coil refeed flash	R9 at risk (15 psi is 1.03 bar)
Chimney outlet, hood outer skin	2.4 m; 23 °C	R10 met
Loaded mass, full tank	536 kg (402 kg with tank and seal drained)	R11 not met
Overall width	1.48 m	R11 at risk
Parts cost	\$1,915	R12 not met (\$1,800 budget)

The main performance gap is heat transfer. One coil in the firebox takes heat mostly by radiation, and the flue gas leaves the economizer at about 450 °C. The economizer cannot help much because 30 kg/h of feed water can absorb only about 3 kW before it boils. Tighter air control and more heat transfer surface in the flue are both needed to reach 65 %; a combustion air preheater alone reaches about 62 %.

The second gap is steam per square meter. Soil behind the steam front sits at 100 °C, not 75 °C, and the front must go 1.5 cm past the working depth to hold 70 °C through the hold. In soils finer than sand, the steam cannot be pushed through 16.5 cm of soil at full flow without lifting a 29 kg hood, so real rates may be lower still.

SteamRoot: cutaway

CONCEPT, NOT FOR FABRICATION

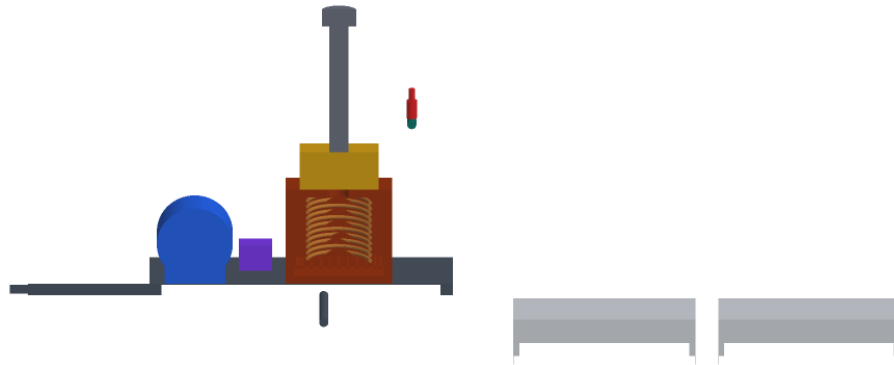


Figure 4. Cutaway showing the helical coil around the fire bed inside the fiber-lined firebox, and the open-bottom hoods with their soil skirts.

■ Key design choices

- **Open-vented, never sealed.** The steam side is always open to the atmosphere through the water-seal pot and vent. There is no isolating valve between the coil and the vent, and the diverter can only send steam to the hose or the vent, never close both. Decided by Amish, 2026-09-25.
- **Monotube coil instead of a drum boiler.** A once-through coil holds under 6 L of water, so a failure releases little stored energy. The cost is careful feed control: too little water dries the coil. Decided by Amish, 2026-09-25.
- **Large-bore stainless coil (25.4 mm, 316).** A 12.7 mm coil would need more than 2 bar at its inlet. The 25.4 mm coil keeps the header near 0.03 bar, though the inlet is at 0.11 bar. Stainless tolerates a dry-fire event far better than copper. Decided by Amish, 2026-09-25.
- **Hood steaming with two hoods used alternately.** Decided by Amish, 2026-09-25. One hose is moved between the hoods with a steam-rated coupling after steam is diverted to the vent. A second hose and a three-outlet diverter would avoid handling the hose but add at least \$180. Proposed: one hose, awaiting Amish.
- **Feed control.** Fixed pump rate with a coil outlet temperature alarm and a tank low-level alarm. Decided by Amish, 2026-09-25.

- **Fiber lining, not castable.** A 50 mm ceramic fiber lining keeps the firebox near 98 kg; castable refractory would make it about 301 kg and break R11. Fiber is less robust against loading damage. Proposed: fiber, awaiting Amish.
- **Wood only.** Split wood or chips at 20 % moisture or drier, batch-fed by hand. No liquid fuel backup. Proposed, awaiting Amish.

■ Safety

SteamRoot is the highest-risk concept in the portfolio. Every later stage must keep this section and extend it.

Safety: regulatory ruling first. A written ruling from the local boiler authority on this open-vented, fired monotube coil is an open prerequisite. No detail design, build or firing may happen before it, and its ruling overrides this document if it is stricter.

Safety: never sealed. The steam side must always be open to the atmosphere through the water-seal vent. Never fit a valve, cap or plug that can isolate the coil or header from the vent. Keep the seal pot filled to its mark; an empty seal pot is an open vent, but an overfilled or frozen one raises the relief pressure. Never plug, adjust or remove the relief valve. A sealed coil or tank over a fire can rupture violently.

Safety: certified relief valve. Fit only a certified, code-stamped steam safety valve sized for the full steam output. It is a second line of protection behind the open vent, not a replacement for it. Test it by its lever as the maker directs.

Safety: local boiler rules. Many jurisdictions regulate steam generators by pressure, heating surface, water volume or firing rate, and some regulate all fired steam equipment. The low-pressure design is intended to stay below common thresholds, but this is unconfirmed. Confirm with the local boiler inspector or authority before any build, and follow their ruling even if it is stricter than this document.

Safety: steam burns. Steam at 100 °C releases its latent heat on skin and causes deep burns within a second. Divert steam to the vent before lifting a hood or moving the hose coupling. In fine soils steam may jet from under the hood skirt, because the soil resists steam flow more than the hood's weight can hold (STR-CAL-001, section 2). Keep hands and feet clear of the skirt and the vent outlet. Wear heat-resistant gloves, a face shield, long sleeves and closed boots. Keep bystanders, children and animals at least 5 m away. Use a steam-rated hose with a whip check, and inspect it before every use.

Safety: dry coil. If the feed pump stops or the tank runs dry, the coil heats toward the firebox gas temperature (about 520 °C) while the fire keeps burning. Refeeding it could flash about 1.3 kg of water at about five times the design steam flow, more than the vent path is sized for. Never restart the feed into a hot, dry coil. Close the air damper, let the fire die and let the coil cool first. The coil outlet alarm and the tank low-level alarm exist to catch this early.

Safety: fire. Operate only on bare ground or gravel, clear of dry vegetation, with a fire extinguisher and water at hand. The spark arrestor must be fitted; the flue leaves at about 450 °C. Do not operate during fire bans or in high wind. Never leave a lit firebox unattended, and let it burn out fully before towing.

Safety: carbon monoxide. Wood fires produce carbon monoxide, which is odorless and deadly. The firebox and chimney must stay outdoors. Steam may be piped into a greenhouse or polytunnel, but the generator may not. The operator wears a personal CO alarm. Never refuel or tend the fire from downwind in the smoke.

Safety: hot surfaces and water quality. The firebox skin reaches about 100 °C, and the chimney, economizer, header, seal pot and hood inner skin reach burn temperatures. Use rain water or softened water to limit scale in the coil; scale raises tube temperature and can block the coil.

- **Soil biology.** Steaming kills beneficial organisms as well as pests, and the soil behind the steam front reaches 100 °C. Overheating soil can release manganese and ammonium that harm seedlings. Do not steam longer than needed.
- **Towing.** Tow only with the fire out, the tank and seal pot drained or secured, and the hoods strapped down. Towing with a full tank puts the loaded mass at about 536 kg.

■ Open questions

- Regulatory status of an open-vented, fired monotube coil in Amish's home jurisdiction. Open prerequisite; gates all further work.
- How can R5 be closed: a convective evaporator bank in the flue, controlled primary and secondary air, a combustion air preheater, or a combination? Awaiting Amish.
- What soil permeability and skirt leakage do real beds show, and does the hood need ballast? This decides R1 and R2 and cannot be settled on paper.
- Where do two hoods ride during towing? The deck behind the firebox is too short for them.
- How is the seal pot level kept up as steam condenses in it and bubbles through it, and how is it protected from freezing?
- How much does starting soil moisture change steam demand? Wet soil needs more energy but conducts heat better.

Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: [STR-DWG-002](#).